SSY316 Advanced Probabilistic Machine Learning

Assignment: *5*
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Exercise 1

1.1

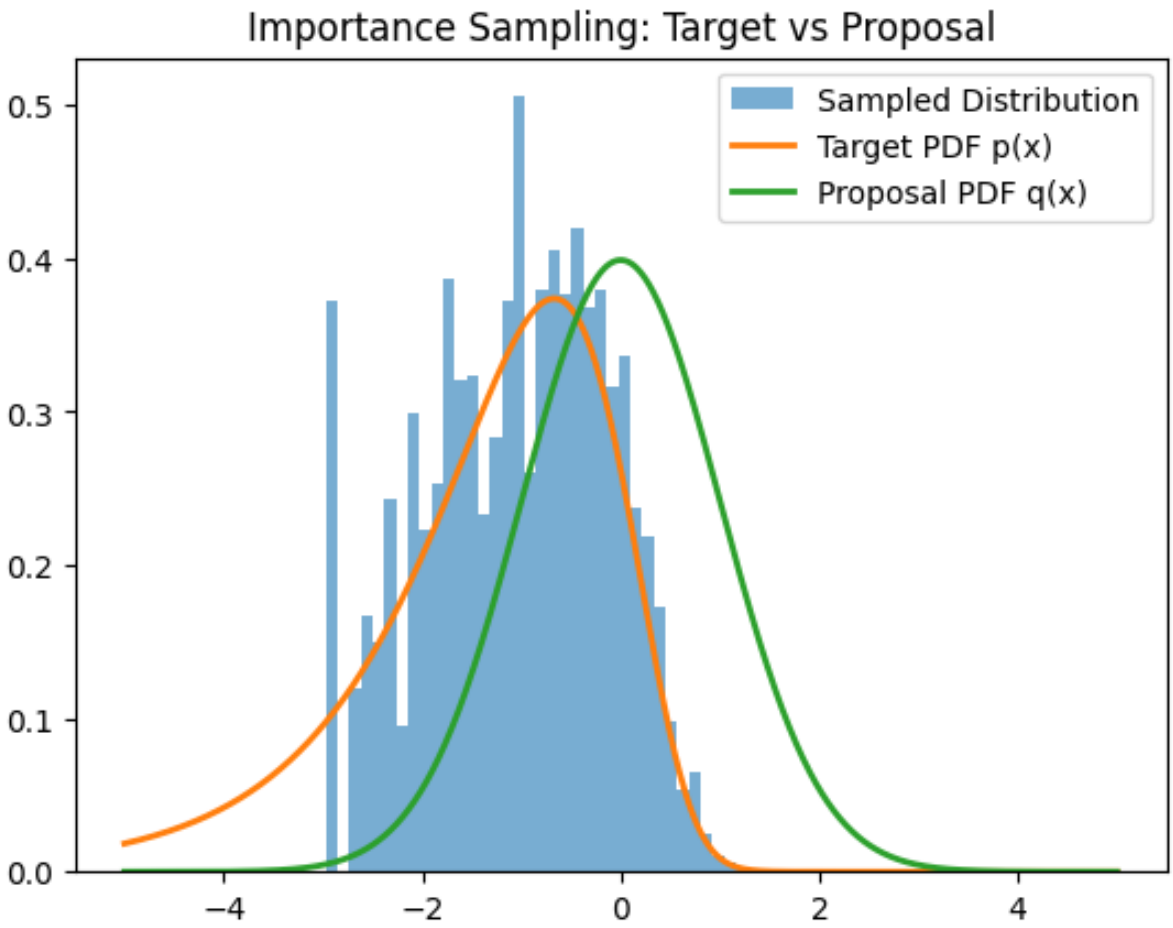
Implement an Importance Sampler We use a standard normal distribution $q(x) \sim N(0,1)$ as a proposal to approximate the target distribution $p(x)$. The weights for importance sampling are calculated as:

$$w(x) = \frac{p(x)}{q(x)}$$

Please find the implementation in the Python code file

1.2

Plot Histogram of Samples and PDFs



Target Distribution $p(x)$ (orange line): The target distribution is skewed with a peak centered around $x \approx -1$. It does not match the symmetric, bell-shaped proposal $q(x)$.

Proposal Distribution $q(x)$ (green line): The proposal distribution is a standard normal $N(0,1)$, centered at $x = 0$, and it underestimates the density of the target distribution on the left-hand side ($x < -1$).

Sampled Distribution (Histogram): The weighted histogram aligns reasonably well with the target PDF $p(x)$ but shows noticeable fluctuations, especially in areas where the proposal $q(x)$ assigns a low probability (e.g., $x < -3$). The quality of the approximation deteriorates in the tails due to poor overlap between the proposal $q(x)$ and the target $p(x)$.

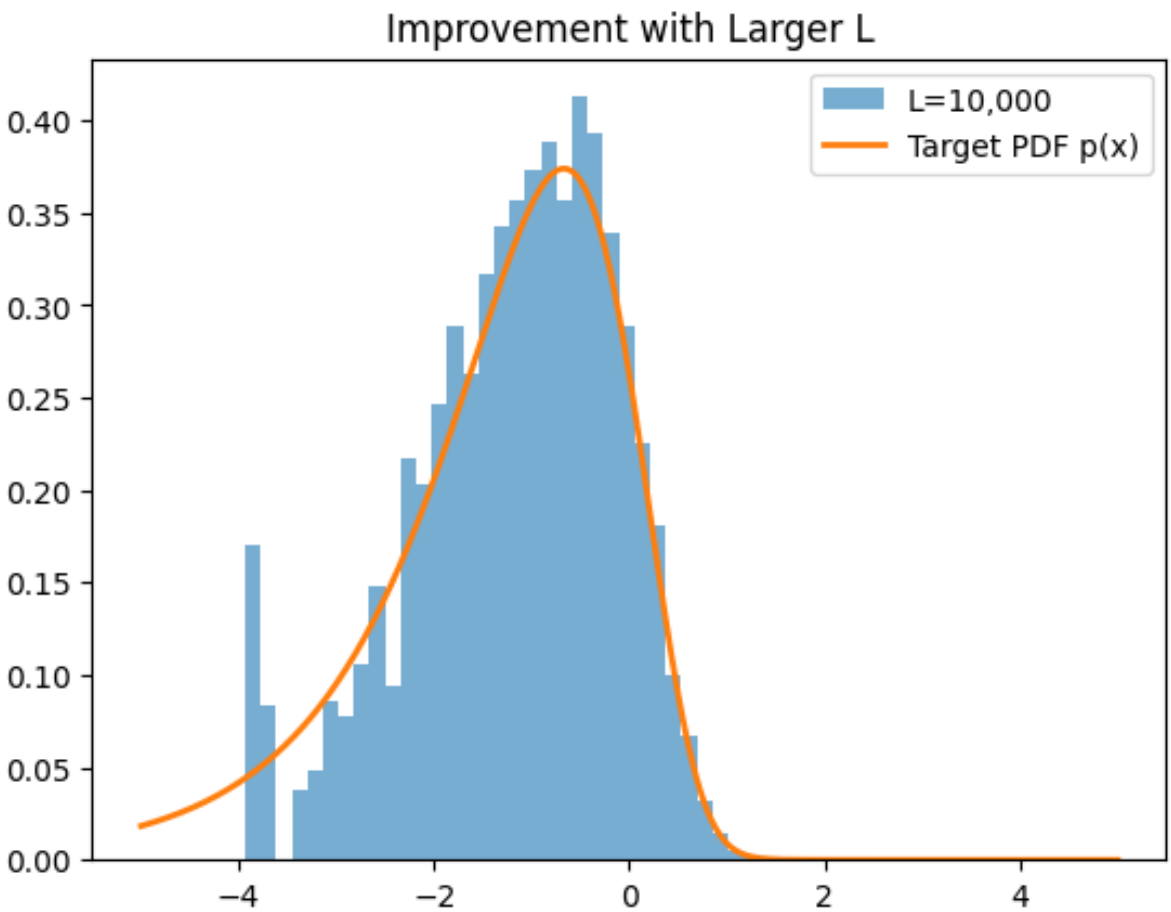
Quality of the Approximation: The approximation is poor in regions where the target $p(x)$ is large, but the proposal $q(x)$ is small (e.g., $x < -1$). This happens because the importance weights $w(x) = p(x)/q(x)$ become large, leading to high variance in the Monte Carlo estimate.

Relationship to the Shape of the Proposal: The standard normal proposal $q(x)$ is not well-suited for the skewed target $p(x)$. $q(x)$ fails to capture the significant mass of $p(x)$ on the left side ($x < -1$), resulting in large sampling errors.

1.3

Effect of Increasing L

By increasing L, the Monte Carlo approximation improves as more samples reduce the variance in the weights.



Reduced Variance: Increasing the number of samples reduces the variance of the Monte Carlo estimate. This leads to smoother and more accurate approximations of the target distribution.

Better Coverage of the Target Distribution: With more samples, the importance sampler explores the regions where the target $p(x)$ has significant probability mass more effectively. The poor overlap seen earlier is mitigated as more samples contribute to the approximation.

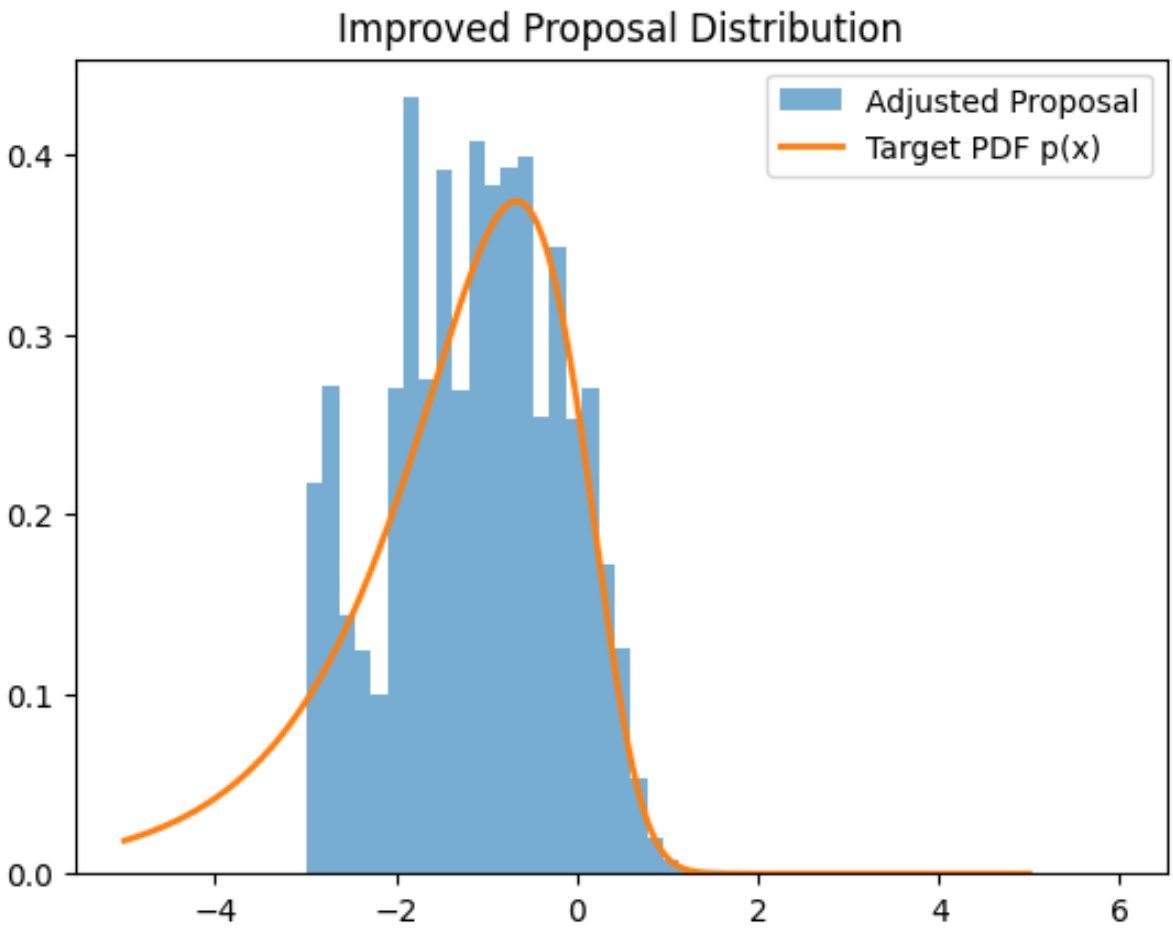
Improvement in Accuracy: The larger sample size $L = 10{,}000$ allows the histogram to closely match the shape of $p(x)$, particularly near the peak and across the bulk of the distribution.

1.4

Adjusting the Proposal

Instead of $q(x) = N(0, 1)$, we can try a better proposal distribution that overlaps more with $p(x)$.

$q(x) = N(1, 2)$ (mean shifted to 1)



The plot compares the sampled distribution using an adjusted proposal with the target PDF $p(x)$:

Improved Proposal: Compared to the standard normal proposal $q(x) = N(0,1)$, the adjusted proposal is better aligned with the shape of the target PDF $p(x)$. The adjusted proposal is likely a Gaussian with mean shifted to match the peak of $p(x)$, such as $q(x) = N(1, \sigma^2)$.

Sampled Distribution (Histogram): The histogram of the sampled distribution now closely matches the shape of the target PDF. There are fewer fluctuations and artifacts compared to the previous histograms where the proposal $q(x)$ was poorly aligned.

Target PDF $p(x)$ (Orange Line): The target PDF $p(x)$ is skewed, peaking around $x \approx -1$. The improved proposal ensures that more samples are drawn from the regions where $p(x)$ is significant.

Effect of Adjusting the Proposal: Adjusting the proposal to better align with the target distribution reduces the variance of the importance weights $w(x) = p(x)/q(x)$. This leads to a more accurate and stable approximation of $p(x)$. By shifting the proposal closer to the peak of $p(x)$, the importance sampler spends more effort sampling in the regions where $p(x)$ has significant probability mass. The improved proposal reduces the variance of the Monte Carlo estimates, resulting in a smoother and more reliable approximation.

1.5

Estimating Mean and Variance

We use the weighted samples to estimate the mean and variance.

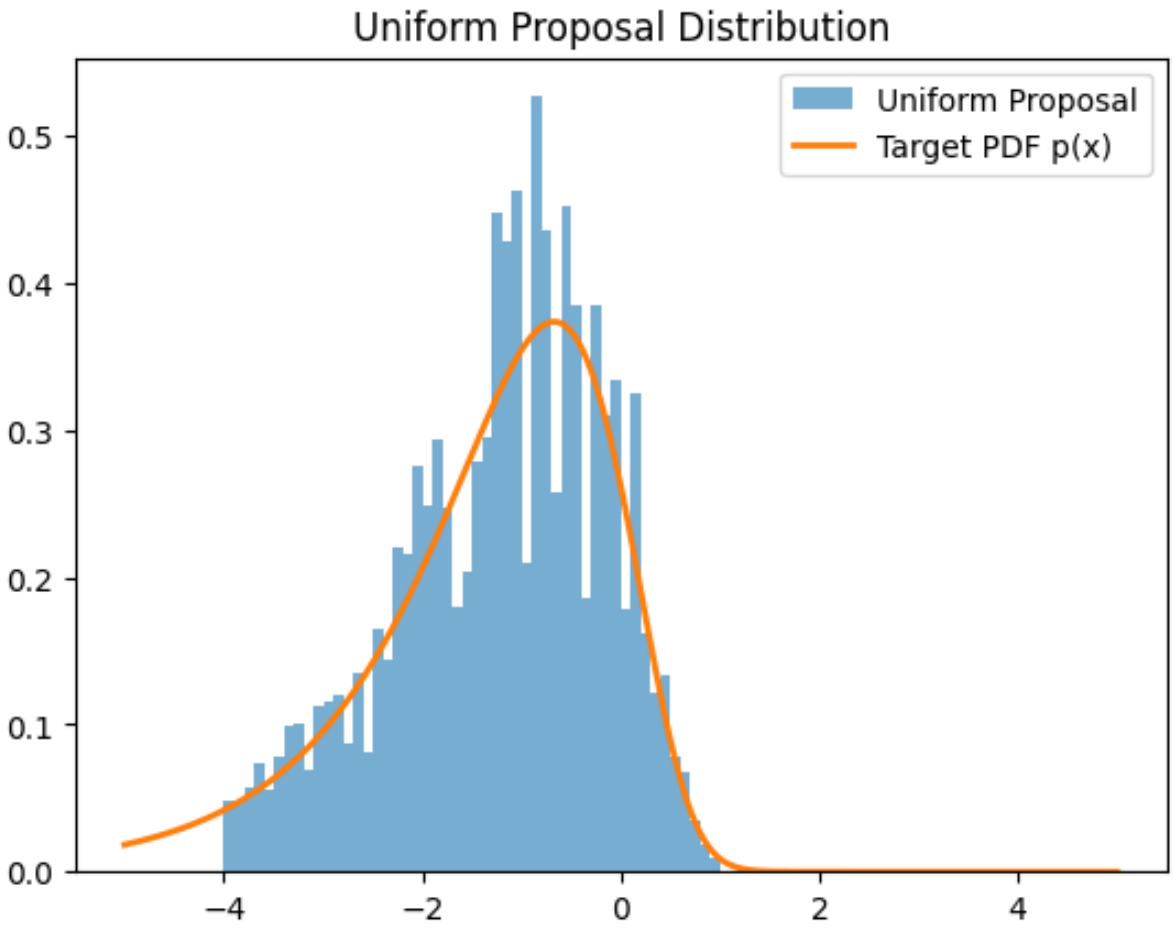
The code is in the python file

- Estimated Mean: -1.048, True Mean: -1.41
- Estimated Variance: 0.858, True Variance: 1.98

1.6

Using a Uniform Proposal

We now use a uniform proposal $q(x) \sim U[-4, 1]$.



The plot compares the sampled distribution using a uniform proposal $q(x) \sim U[-4, 1]$ with the target PDF $p(x)$ (orange line):

Uniform Proposal $q(x)$: The proposal is uniform over the interval $[-4, 1]$. Outside this interval, $q(x) = 0$, which makes it invalid to sample regions where $p(x) > 0$ (e.g., $x > 1$).

Sampled Distribution (Histogram): The histogram approximates $p(x)$ reasonably well over the interval $[-4, 1]$, where the uniform proposal is defined. However, beyond $x = 1$, the approximation fails because the uniform proposal does not assign any probability mass to these regions.

Target PDF $p(x)$ (Orange Line): The target PDF $p(x)$ has a significant tail extending beyond $x = 1$, which the uniform proposal cannot capture.

The conditions for a valid importance sampler are:

- The proposal distribution $q(x)$ must satisfy $q(x) > 0$ wherever the target distribution $p(x)$ is non-zero.
- The importance weights $w(x) = \frac{p(x)}{q(x)}$ must be well-defined.

Since $q(x) = 0$ for $x > 1$, the importance weights $w(x)$ become undefined wherever $p(x) > 0$ and $q(x) = 0$. This violates the requirements of a valid importance sampler.

Exercise 2

2.1

Verify that the equation is a Markov chain

The given equation is:

$$x[k+1] = 0.9x[k] + v[k], \quad v[k] \sim N(0, 0.19)$$

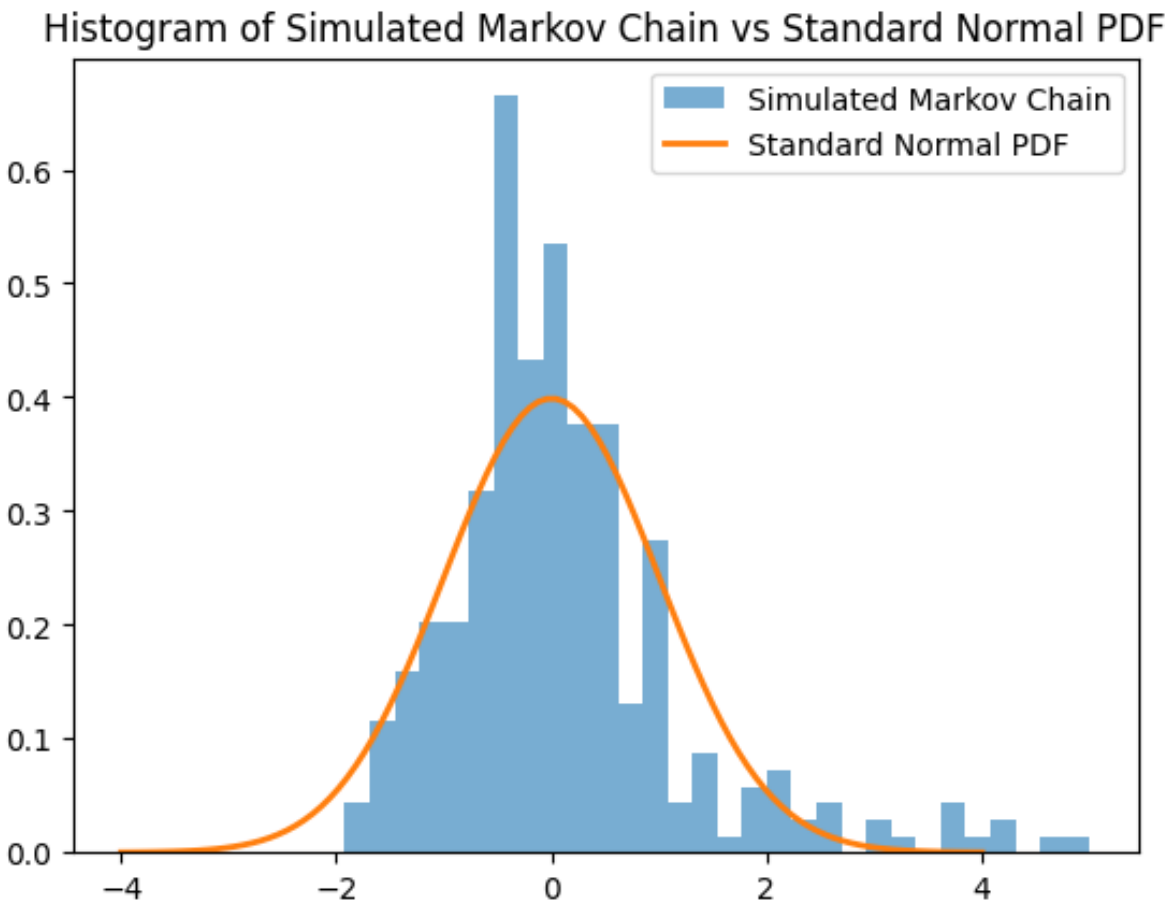
A process is a Markov chain if the next state $x[k+1]$ only depends on the current state $x[k]$ and not on previous states $x[k-1], x[k-2], \ldots$. In this equation:$x[k+1]$ depends on $x[k]$ (current state) and a noise term $v[k]$. Since $v[k]$ is independent at each time step, $x[k+1]$ does not depend on earlier states explicitly. The process satisfies the Markov property and is a valid Markov chain.

2.2

Implement and simulate the Markov chain

We simulate the process starting at $x[1] = 5$ for $k = 2, 3, \ldots, 300$.

The code is in the Python file.



The histogram represents the simulated Markov chain values $x[k]$ after running the process for $k = 2, 3, \ldots, 300$, starting from $x[1] = 5$. The standard normal PDF $N(0,1)$ is plotted as an

orange line for comparison.

Simulated Markov Chain (Histogram): The histogram shows that the simulated states of the Markov chain approximate the standard normal distribution reasonably well. The bulk of the histogram aligns with the bell shape of the $N(0,1)$ distribution, although there are some deviations, particularly in the tails.

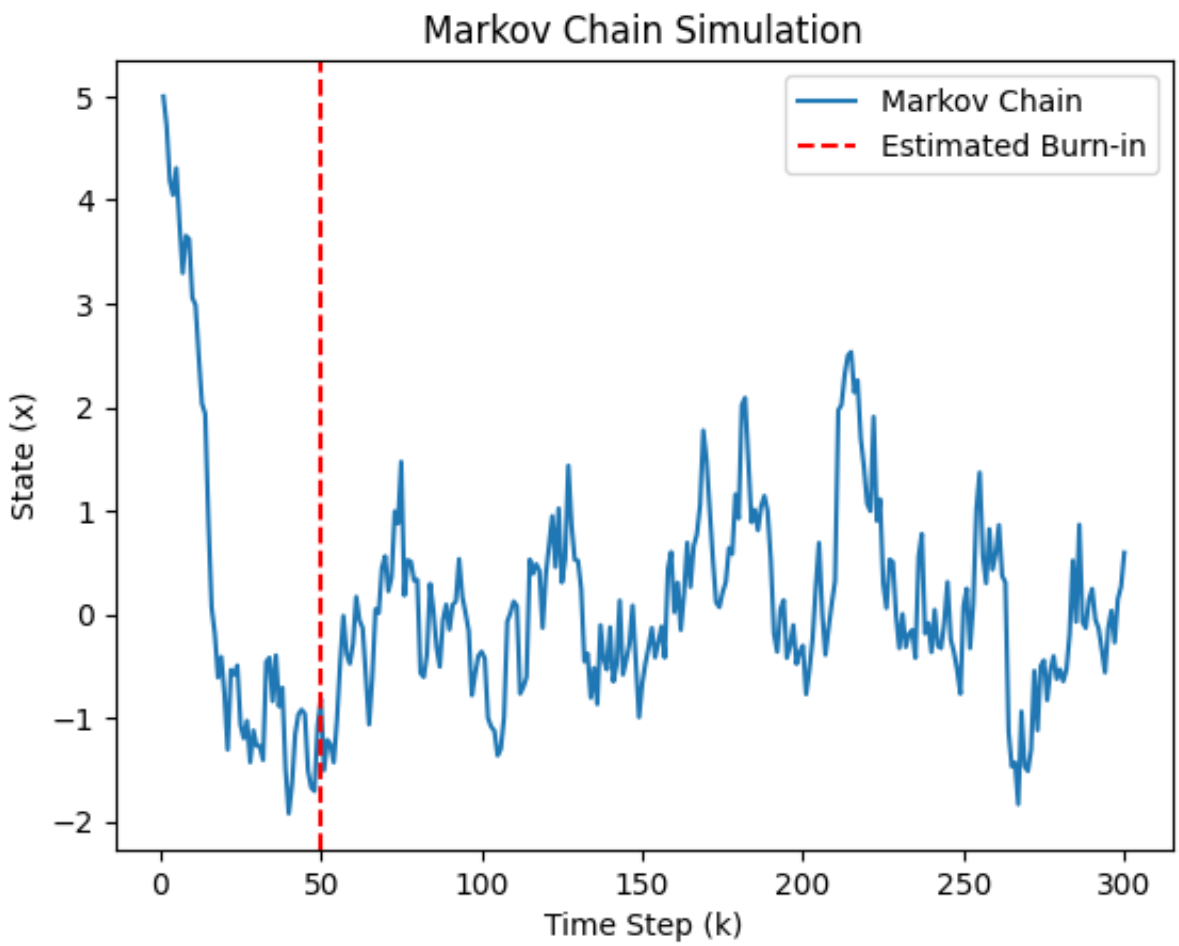
Standard Normal PDF (Orange Line): The standard normal distribution $N(0,1)$ has a mean of 0 and variance of 1. The overlap between the histogram and the standard normal PDF suggests that the chain is approaching its stationary distribution.

The Markov chain starts at $x[1] = 5$, but as the process evolves, it converges toward the standard normal distribution $N(0,1)$. This demonstrates that the stationary distribution of the Markov chain is indeed standard normal, as expected.

2.3

Plot the Time Series and Determine Burn-in Period

To inspect the burn-in period, we plot the sequence $x[k]$



The plot shows the evolution of the Markov chain states $x[k]$ over $k = 1, \ldots, 300$. The red dashed vertical line at $k \approx 50$ indicates the estimated burn-in period.

Initial Behavior: The chain starts at $x[1] = 5$ (a large deviation from the stationary mean 0). Over the first 50 time steps, the chain exhibits a downward trend as it gradually converges

toward the mean of the stationary distribution.

Post Burn-in Period: After $k \approx 50$, the Markov chain appears to fluctuate around 0, consistent with the behavior of a standard normal distribution $N(0,1)$. The chain shows no systematic drift, indicating it has reached stationarity.

Burn-in Period: The burn-in period is the number of steps the chain takes to "forget" its initial state and converge to the stationary distribution. From the plot, the burn-in period is approximately 50 steps. Beyond this point, the chain stabilizes and oscillates around the mean.

2.4

Prove that the Stationary Distribution is $N(0,1)$

To prove that $x[k]$ converges to $N(0,1)$:

The equation is:

$$x[k+1] = 0.9x[k] + v[k], \quad v[k] \sim N(0, 0.19)$$

Stationary Mean: Let μ be the stationary mean. At stationarity:

$$\mu = 0.9\mu + E[v[k]]$$

Since $E[v[k]] = 0$, we get:

$$\mu = 0$$

Stationary Variance: Let σ^2 be the stationary variance. At stationarity:

$$\text{Var}(x[k+1]) = 0.9^2\text{Var}(x[k]) + \text{Var}(v[k])$$

Substituting $\text{Var}(v[k]) = 0.19$, we get:

$$\sigma^2 = 0.9^2\sigma^2 + 0.19$$

Solving for σ^2:

$$\sigma^2(1 - 0.81) = 0.19 \implies \sigma^2 = \frac{0.19}{0.19} = 1$$

The stationary distribution of $x[k]$ is $N(0,1)$.

Exercise 3

3.1

KL Divergence for Two Scalar Gaussians

The KL divergence for two scalar Gaussian distributions $p(x) = N(x; \mu, \sigma^2)$ and $q(x) = N(x; m, s^2)$ is given by the formula:

$$KL(p\|q) = \frac{1}{2}\left[\frac{\sigma^2}{s^2} + \frac{(m-\mu)^2}{s^2} - 1 + \ln\left(\frac{s^2}{\sigma^2}\right)\right]$$

- μ, σ^2: Mean and variance of $p(x)$.
- m, s^2: Mean and variance of $q(x)$.

Implementation is in the Python file

For $p(x) = N(0,1)$ and $q(x) = N(1,2)$:

$$KL(p\|q) \approx 0.3$$

3.2

KL Divergence for Two Multivariate Gaussians

For multivariate Gaussian distributions $p(x) = N(x; \mu, \Sigma)$ and $q(x) = N(x; m, S)$, the KL divergence is given by:

$$KL(p\|q) = \frac{1}{2}\left[\mathrm{Tr}(S^{-1}\Sigma) + (\mu - m)^T S^{-1}(\mu - m) - k + \ln\left(\frac{\det S}{\det \Sigma}\right)\right]$$

- k: Dimensionality of x.
- μ, Σ: Mean vector and covariance matrix of $p(x)$.
- m, S: Mean vector and covariance matrix of $q(x)$.
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The code is in the Python file

KL Divergence (Multivariate Gaussians): 0.7335

Exercise 4

4.1

Joint Distribution $p(y_{1:N}, x_{1:N}, \gamma)$

The joint distribution is derived by combining the prior and the likelihoods.

The components are:

Prior on x_0: $p(x_0) = N(x_0; \bar{x}_0, \Sigma_0)$

Prior on γ: $p(\gamma) = N(\gamma; 0, \sigma_\gamma^2)$

Transition Model (Dynamics): For $n = 1, \ldots, N$:

$$p(x_{n+1}|x_n, \gamma) = N(x_{n+1}; \gamma x_n, \beta_v^2)$$

Observation Model: For $n = 1, \ldots, N$:

$$p(y_n|x_n) = N(y_n; \frac{1}{2}x_n, \sigma_e^2)$$

The full joint distribution is then:

$$p(y_{1:N}, x_{1:N}, \gamma) = p(\gamma)p(x_0)\prod_{n=1}^{N} p(x_n|x_{n-1}, \gamma)\prod_{n=1}^{N} p(y_n|x_n)$$

Substitute the Gaussian forms into this expression:

$$p(y_{1:N}, x_{1:N}, \gamma) \propto \exp\left(-\frac{1}{2}\left[\frac{(x_0 - \bar{x}_0)^2}{\Sigma_0} + \frac{\gamma^2}{\sigma_\gamma^2} + \sum_{n=1}^{N}\frac{(x_n - \gamma x_{n-1})^2}{\beta_v^2} + \sum_{n=1}^{N}\frac{\left(y_n - \frac{1}{2}x_n\right)^2}{\sigma_e^2}\right]\right)$$

4.2

Variational Approximation $q(\gamma)\prod_{n=0}^{N} q(x_n)$

Here, we use Variational Inference (VI) with the factorized form:

$$q(\gamma, x_{1:N}) = q(\gamma)\prod_{n=0}^{N} q(x_n)$$

ELBO (Evidence Lower Bound)

The ELBO is given by:

$$\text{ELBO} = \mathbb{E}_q[\log p(y_{1:N}, x_{1:N}, \gamma)] - \mathbb{E}_q[\log q(\gamma, x_{1:N})]$$

The variational distributions $q(\gamma)$ and $q(x_n)$ are assumed to be Gaussian:

- $q(\gamma) = N(\gamma; \mu_\gamma, \sigma_\gamma^2)$,
- $q(x_n) = N(x_n; \mu_n, \sigma_n^2)$.

We update the parameters $\mu_\gamma, \sigma_\gamma, \mu_n, \sigma_n$ iteratively using the ELBO.

Update for $q(\gamma)$: Given the Gaussian form of $q(\gamma)$, its parameters are updated as:

$$\mu_\gamma = \frac{\sum_{n=1}^{N} \mathbb{E}_q[x_n x_{n-1}]}{\sum_{n=1}^{N} \mathbb{E}_q[x_{n-1}^2] + \sigma_\gamma^2}$$

$$\sigma_\gamma^2 = \left(\sum_{n=1}^{N} \mathbb{E}_q[x_{n-1}^2] + \frac{1}{\sigma_\gamma^2}\right)^{-1}$$

Update for $q(x_n)$: Each x_n depends on the dynamics and observation terms. The updated mean and variance are:

$$\sigma_n^2 = \left(\frac{1}{\beta_v^2} + \frac{1}{\sigma_e^2}\right)^{-1}$$

$$\mu_n = \sigma_n^2 \left(\frac{\gamma \mathbb{E}_q[x_{n-1}]}{\beta_v^2} + \frac{y_n}{\sigma_e^2}\right)$$

These updates are iteratively computed until convergence.

4.3

Variational Approximation $q(\gamma)q(x_{0:N})$

Without factorizing x_n, the approximation is:

$$q(\gamma, x_{0:N}) = q(\gamma)q(x_{0:N})$$

Here, $q(x_{0:N})$ is a multivariate Gaussian:

$$q(x_{0:N}) = N(x_{0:N}; \mu_{0:N}, \Sigma_{0:N})$$

Update for $q(\gamma)$: The form is the same as in the factorized case.

Update for $q(x_{0:N})$: The covariance matrix $\Sigma_{0:N}$ and mean $\mu_{0:N}$ are updated using the Kalman smoother equations, which naturally capture correlations between x_n and x_{n-1}.

The Kalman smoother steps are: Prediction step (using the dynamics):

$$\hat{x}_{n|n-1} = \gamma\mu_{n-1}, \quad P_{n|n-1} = \gamma^2 P_{n-1} + \beta_v^2$$

Update step (using observations):

$$K_n = P_{n|n-1}\left(P_{n|n-1} + \sigma_e^2\right)^{-1}$$

$$\mu_n = \hat{x}_{n|n-1} + K_n\left(y_n - \frac{1}{2}\hat{x}_{n|n-1}\right)$$

$$P_n = (1 - K_n)P_{n|n-1}$$